Ex-Ante Motor Insurance Premium Prediction Using Machine Learning

A Leakage-Aware and Interpretable Pricing-Time Modeling Framework

Abstract

Motor insurance premium prediction is a complicated and also important business-sensitive pricing problem. A reliable pricing model must only using the information available at the time before contract renewal in order to avoid future information leakage. Handling imperfect insurance data and providing interpretable explanations for its predictions are also important. This study analyzes a Spanish motor insurance policy dataset with 105,555 annual policy observations with rigorous statistical machine learning workflow and develops an ex-ante framework for predicting motor insurance premiums.

To completely maintain the ex-ante design, variables that may contain post-renewal outcomes, such as claim history, lapse outcomes, and lapse dates, are excluded or treated cautiously. For the remaining variables, this study implemented a scrupulous sanity check procedure, making sure they are audited, cleaned, and transformed through a business-driven feature engineering process. The engineered features include driver age at renewal, license tenure, vehicle age, log vehicle value, power-to-weight ratio, value per vehicle year, policy tenure, renewal timing variables, and abnormality indicators for special or missing vehicle information.

This study used a chronological data splitting method when dealing with time series data. This design reflects the real insurance setting in which models are trained on historical policies for predicting future renewal premiums. The modeling strategies are designed differently for linear models and tree-based models when selecting features and used both raw premium and log-transformed premium targets. Benchmark results indicate that tree-based models outperform linear models, suggesting that premium formation contains nonlinear relationships and interactions among vehicle, customer, and contract variables. After chronological cross-validation and hyperparameter tuning, the final model selecte for this study is a Random Forest trained on log1p(Premium). On the out-of-time test set, the model achieves a MAE of 55.08, RMSE of 91.05, and R^2 of 0.573.

This study also used permutation importance and SHAP analysis to analyze feature importances for better interpretations. The results show that features like value per vehicle year, payment method, abnormal door-count indicator, policy tenure, vehicle value, licence tenure, distribution channel, and vehicle structure are important drivers of premium prediction. These findings suggest that meaningful pricing-related patterns are discovered through models and there's a distinguished improvement compare to simple arbitrary statistical correlations analysis.

Overall, this study provides a reproducible and interpretable ex-ante premium prediction framework that can support insurance pricing analysis while maintaining methodological discipline and business relevance.

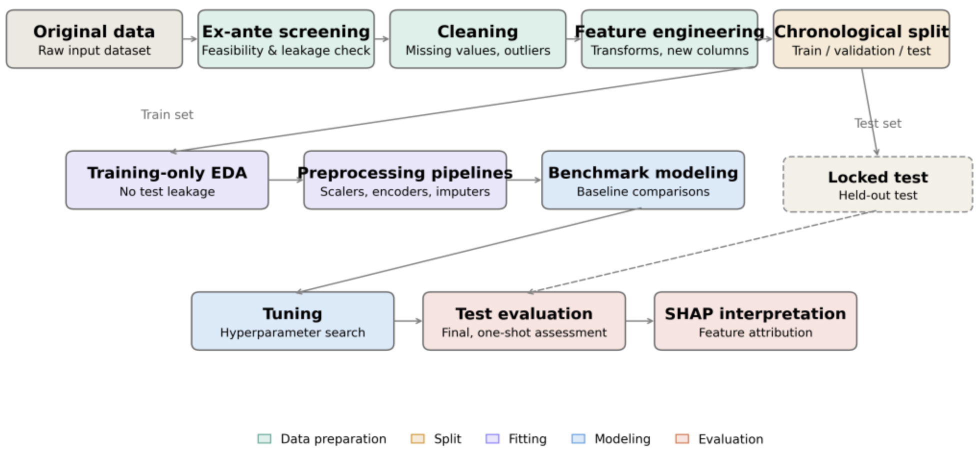
Keywords: motor insurance, premium prediction, ex-ante modeling, feature engineering, chronological validation, Random Forest, SHAP, interpretability

1. Introduction

This study develops a comprehensive ex-ante machine learning workflow for motor insurance premium prediction. The workflow begins with understanding the variables and auditing the data quality because insurance variables have business meaning and timing constraints. Then it does rule-based cleaning, builds business-driven features, splits data chronologically, runs training-only exploratory data analysis, builds model-specific preprocessing pipelines, compares benchmark models, tunes candidate models with chronological validation, evaluates final model with out-of-time test set, and interprets final model with feature importance and SHAP.

Thus, this project provides a method and a prediction. It illustrates how a machine learning premium prediction model can be built under realistic constraints of pricing-time information, and how its results can be interpreted in terms of insurance business logic.

Figure 1. Overall Project Workflow



2. Data Source and Variable Understanding

2.1 Original Dataset

The data of this study comes from a Spanish car insurance company, it contains 105,555 annual policy observations and 53,502 unique policy IDs. The target variable is Premium, which represents the observed policy premium. The policy renewal dates range from 2015-11-02 to 2018-11-30.

Table 1. Original variable groups

Variable Group	Examples	Role in the Study
Identification and time variables	ID; Date_start_contract; Date_last_renewal; Date_next_renewal; Date_lapse	Define the policy timeline and renewal reference date
Customer and contract variables	Seniority; Policies_in_force; Max_policies; Max_products; Payment; Distribution_channel; Second_driver; Area; Type_risk	Describe customer relationship, contract structure, channel, and risk classification
Customer date variables	Date_birth; Date_driving_licence	Used to construct age and driving experience at renewal
Vehicle variables	Power; Cylinder_capacity; Value_vehicle; N_doors; Type_fuel; Length; Weight; Year_matriculation	Describe vehicle value, performance, size, structure, and age
Claim and lapse variables	Claim counts; claim costs; lapse indicator; lapse date	Excluded or treated cautiously because of potential post-renewal information
Target variable	Premium	Observed policy premium to be predicted

2.2 Variable-Level Understandings

A key part of this project is inspecting variables before modeling. This step is especially important for insurance data because each variable represents a business concept, not just a statistical feature.

Since the model is designed for an ex-ante premium prediction, so all variables must be judged by whether they are observable at the renewal pricing date. Some variables may be predictive but invalid because they describe events occurring after pricing. Moreover, insurance datasets often contain business-coded categories, missing values, special codes, and physically unreasonable values. For example, a value of zero for vehicle doors or missing vehicle power may not be a simple random error, it may indicate a special vehicle record, an incomplete administrative entry, or an unusual underwriting case. So this study enhance data quality by a rigorous sanity check.

In this project, variable inspection is not treated as a minor preprocessing step. It is part of the modeling design because it determines the valid information set, the cleaning logic, and the engineered feature space.

2.3 Ex-Ante Feature Screening

The ex-ante screening stage defines which variables can legitimately be used for premium prediction. The guiding rule is that a variable should only be used if it would reasonably be known at the time of pricing or renewal.

Customer information, vehicle information, contract structure, and renewal-time variables are retained because they are available before or at the pricing decision. In contrast, current-year claim outcomes, claim costs, lapse outcomes, and lapse dates are excluded from the main model because they may occur after the renewal premium has already been set. Variables with ambiguous timing, such as full-life claim summaries, are treated cautiously because they may include information beyond the pricing date.

This conservative screening rule is important. Excluding potentially leaky variables may reduce raw predictive power, but it makes the evaluation more realistic and prevents the model from learning future outcomes. The resulting model is therefore closer to what could be used in a real renewal pricing environment.

Table 2. Ex-ante variable screening rules

Variable Type	Main Decision	Reason
Customer profile available at renewal	Used	Known before or at the renewal pricing date
Contract structure available at renewal	Used	Describes current policy relationship and payment/channel information
Vehicle attributes	Used	Vehicle information is part of pricing-time underwriting data
Date variables	Used after transformation	Raw dates are converted into renewal-time age, tenure, and vehicle age
Current-year claim count	Excluded	May describe events after the pricing decision
Current-year claim cost	Excluded	Post-renewal outcome and not available at pricing time
Lapse indicator	Excluded	Represents future policy cancellation outcome
Lapse date	Excluded	Not known unless lapse has already occurred
Ambiguous full-life claim summaries	Excluded or treated cautiously	May include future information if not strictly lagged

3. Methodology

The methodology of this study is designed to support a realistic premium prediction setting. The main stages are data quality auditing, rule-based cleaning, business-driven feature engineering, chronological data splitting, training-only exploratory analysis, and preprocessing pipeline construction. These stages are connected by one central goal: building a model that is predictive, leakage-free, and interpretable.

3.1 Data Quality Audit and Rule-Based Cleaning

The first methodological stage is a systematic audit of the raw data. The audit focuses on four types of problems: inconsistent categorical coding, pseudo-missing values, invalid numerical values, and business-logic conflicts.

For categorical variables, the project standardizes coding formats and converts pseudo-missing values into true missing values. Categorical encoding treats different strings as different levels, if the same business category is recorded in multiple formats, the model may learn artificial distinctions. Variables such as Distribution_channel and Type_fuel are necessary to go through this kind of standardization.

For numerical vehicle variables, physically unreasonable values are inspected. A value such as zero vehicle power is not treated as a valid measurement. Instead, invalid values are converted to missing values and handled later through imputation. At the same time, missingness itself may carry information. Therefore, the project creates abnormality indicators such as flag_power_missing, flag_type_fuel_missing, flag_length_missing, and flag_n_doors_zero.

For contract variables, logical consistency is checked. For example, the historical maximum number of policies should not be smaller than the number of policies currently in force. If such contradictions occur, the inconsistent value is treated as invalid rather than being passed directly into the model.

This rule-based cleaning process reflects the nature of insurance data. The purpose is not only to remove technical errors, but also to preserve meaningful abnormal patterns. Missing or unusual records may represent special vehicle types, legacy administrative rules, or business-specific coding practices. By combining invalid-value correction with abnormality flags, the model can use this information without treating invalid measurements as ordinary values.

Table 3. Data quality issues and treatment rules

Issue Type	Example Variables	Treatment	Modeling Reason
Inconsistent category coding	Distribution_channel; Type_fuel	Standardize labels and convert pseudo-missing values to missing	Avoid artificial category fragmentation after one-hot encoding
Invalid vehicle measurements	Power; Length	Convert invalid values to missing	Prevent physically invalid values from being treated as real measurements
Special vehicle pattern	N_doors	Create flag_n_doors_zero	Preserve signal from unusual vehicle records
Missing vehicle information	Power; Type_fuel; Length	Create missingness flags	Allow model to learn whether missingness is informative
Contract logic conflict	Max_policies; Policies_in_force	Treat inconsistent values as missing	Enforce basic business consistency
Licence timing problem	Date_birth; Date_driving_licence	Create licence-tenure flags and handle invalid derived values	Avoid invalid driving-experience calculations

3.2 Business-Driven Feature Engineering

Feature engineering is one of the vital parts of this study. The purpose is to transform raw insurance fields into variables that better represent pricing logic. The engineered features are deterministic and interpretable; they are not created through automatic black-box feature generation.

The first group of engineered features describes the driver. age_at_renewal measures the policyholder's age at the renewal date. licence_tenure measures the number of years between obtaining a driving licence and the renewal date. age_when_licensed captures the age at which the customer received the licence. These variables are more meaningful than raw birth date or licence date because insurance pricing depends on the customer's status at the pricing point.

The second group describes vehicle age and vehicle value. vehicle_age is calculated from the renewal year and the vehicle matriculation year. vehicle_value_log is created to reduce the effect of extreme high-value vehicles. value_per_vehicle_year combines vehicle value with vehicle age and provides a measure of value intensity after accounting for depreciation.

The third group describes vehicle intensity and mechanical structure. power_weight_ratio measures vehicle power relative to weight, while engine_cc_per_power measures engine displacement relative to power. These ratio features are designed to capture vehicle characteristics that may not be fully represented by raw power, weight, or cylinder capacity alone.

The fourth group describes the customer-policy relationship. policy_tenure captures the time between contract start and renewal. policy_headroom compares the historical maximum number of policies with the current number of policies in force. These variables represent the customer's relationship with the insurer and the structure of the policy portfolio.

The fifth group includes renewal timing variables and binary flags. Renewal year, month, and quarter capture time-related pricing patterns. Flags such as flag_young_driver, flag_senior_driver, flag_newly_licensed, and vehicle missingness indicators capture threshold effects and special records.

The final modeling dataset contains 41 features. This feature set is intentionally built from pricing-time customer, contract, and vehicle information. It does not rely on post-renewal outcomes or future claim information.

Table 4. Engineered feature groups

Feature Group	Example Features	Construction Logic	Business Meaning
Driver profile	age_at_renewal; licence_tenure; age_when_licensed	Differences between renewal date, birth date, and licence date	Driver age, maturity, and experience at pricing time
Vehicle age and value	vehicle_age; vehicle_value_log; value_per_vehicle_year	Vehicle age from matriculation year; log value; value divided by vehicle age plus one	Vehicle depreciation, insured value, and value intensity
Vehicle intensity	power_weight_ratio; engine_cc_per_power	Ratios using power, weight, and cylinder capacity	Relative vehicle performance and mechanical structure
Contract relationship	policy_tenure; policy_headroom	Contract duration and difference between maximum and current policy count	Customer relationship and policy portfolio structure
Renewal timing	renewal_year; renewal_month; renewal_quarter	Extracted from last renewal date	Time pattern in renewal pricing
Risk and abnormality flags	flag_young_driver; flag_senior_driver; flag_newly_licensed; flag_n_doors_zero; flag_power_missing; flag_length_missing	Binary indicators based on thresholds or abnormal values	Nonlinear risk groups and special data-quality patterns

3.3 Final Modeling Feature Set

After ex-ante screening, cleaning, and feature engineering, the final model uses 41 features. These include original pricing-time variables, engineered numerical features, categorical variables, and binary indicators.

Table 5. Final feature set

Feature Type	Features
Categorical features	Distribution_channel; Payment; Type_risk; Area; Second_driver; Type_fuel
Original numerical and discrete features	Seniority; Policies_in_force; Max_policies; Max_products; Year_matriculation; Power; Cylinder_capacity; Value_vehicle; N_doors; Length; Weight
Driver and policy features	age_at_renewal; licence_tenure; age_when_licensed; policy_tenure; policy_headroom
Vehicle engineered features	vehicle_age; vehicle_value_log; power_weight_ratio; engine_cc_per_power; value_per_vehicle_year; weight_log; length_log
Renewal timing features	renewal_year; renewal_month; renewal_quarter
Abnormality and risk flags	flag_n_doors_zero; flag_power_missing; flag_type_fuel_missing; flag_length_missing; flag_negative_licence_tenure; flag_licence_tenure_missing; flag_young_driver; flag_senior_driver; flag_newly_licensed

This feature set reflects the specific design of this project. It is not based on previous-year premium anchoring or route-specific renewal modeling. Instead, it estimates premium from general customer, contract, and vehicle information available at renewal time.

3.4 Chronological Data Splitting

The dataset is split chronologically according to Date_last_renewal. This design is necessary because the real pricing problem is forward-looking. In practice, a model is trained using past policies and applied to later renewal periods. A random split would mix earlier and later observations, which may lead to overly optimistic evaluation.

The chronological split creates three datasets: training, validation, and test. The training set is used for model fitting and exploratory analysis. The validation set is used for benchmark comparison, tuning, and model selection. The test set is reserved for final out-of-time evaluation.

Table 6. Chronological data split

Dataset	Rows	Unique IDs	Unique Renewal Dates	Renewal Date Range
Training	58,775	38,715	674	2015-11-02 to 2017-09-06
Validation	21,586	21,586	225	2017-09-07 to 2018-04-19
Test	25,194	25,194	225	2018-04-20 to 2018-11-30

Although some policy IDs appear across multiple periods, the split is intentionally based on time rather than random grouping. The research question is whether historical data can support prediction for future renewal periods. Therefore, preserving the chronological structure is more important than constructing a random policy-level split.

3.5 Training-Only Exploratory Data Analysis

Exploratory data analysis is conducted only on the training set. This prevents validation and test information from influencing the modeling design. The purpose of EDA is to understand the target distribution, missingness structure, feature-target relationships, categorical differences, and redundancy among features.

The target variable Premium is right-skewed. In the training set, the average premium is 317.76, the median is 295.24, and the maximum reaches 2797.51. Most policies are concentrated in a moderate premium range, while a small number of high-premium policies create a long right tail. This pattern motivates testing log1p(Premium) as an alternative target transformation during modeling.

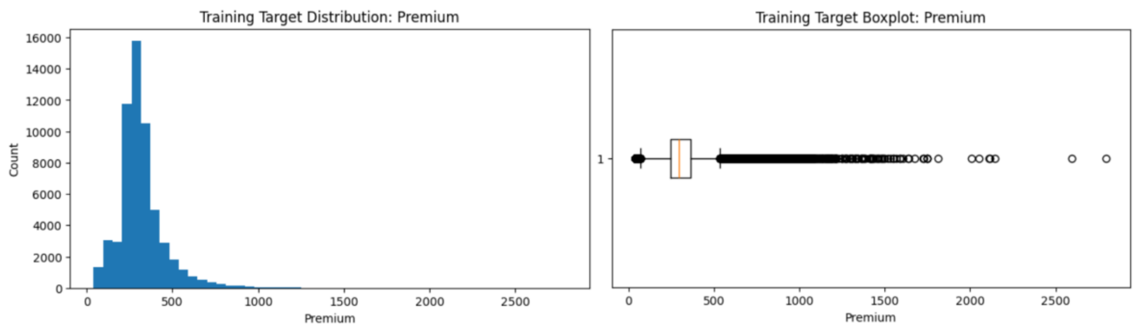


Table 7. Training-set premium summary

Statistic	Value
Count	58,775
Mean	317.76
Standard deviation	141.02
Minimum	40.14
25th percentile	245.19
Median	295.24
75th percentile	362.01
Maximum	2797.51

The missingness analysis shows that missing values are concentrated in a limited number of variables. Length and length_log have the highest missing rate at approximately 9.85%. Distribution_channel has a missing rate of about 3.30%, while Type_fuel, Power, power_weight_ratio, and engine_cc_per_power each have missing rates of about 1.58%. This supports the use of systematic imputation and missingness flags.

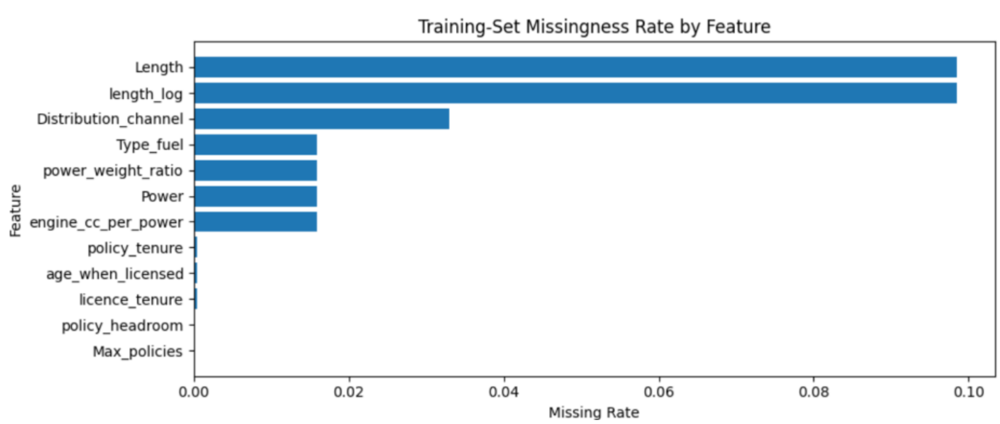


Table 8. Main missingness patterns

Feature	Missing Rate
Length	9.85%
length_log	9.85%
Distribution_channel	3.30%
Type_fuel	1.58%
Power	1.58%
power_weight_ratio	1.58%
engine_cc_per_power	1.58%
policy_tenure	0.04%
age_when_licensed	0.04%
licence_tenure	0.04%

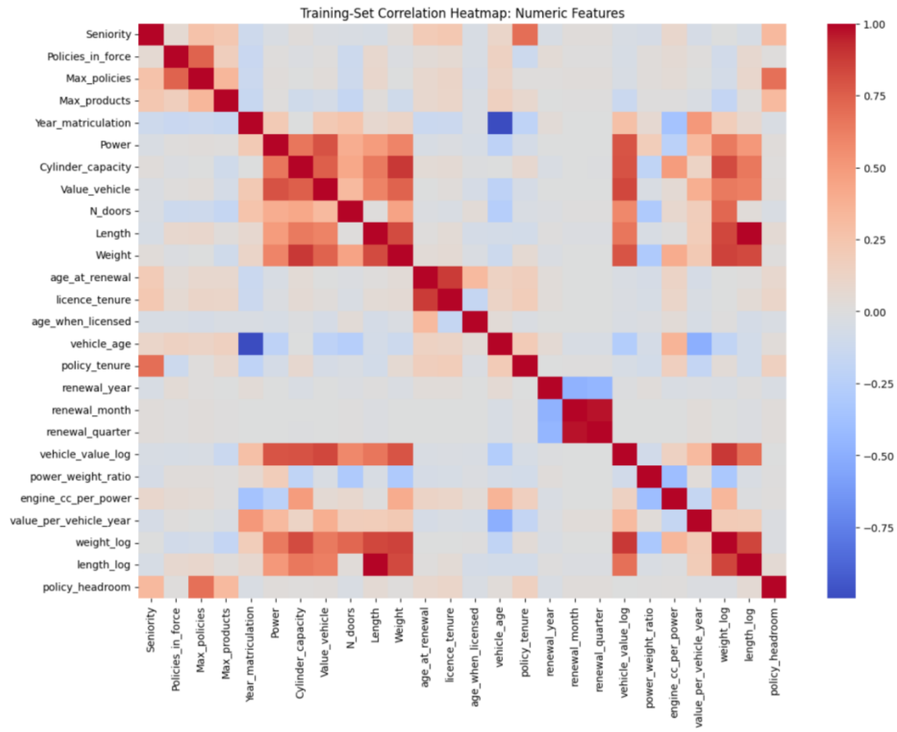
Correlation analysis shows that vehicle-related variables are strongly associated with premium. The highest correlations with premium include vehicle_value_log, Power, Value_vehicle, weight_log, value_per_vehicle_year, and N_doors. vehicle_age is negatively correlated with premium, which is consistent with the idea that newer or more valuable vehicles tend to have higher premiums.

	feature	pearson_corr_with_target
0	vehicle_value_log	0.456907
1	Power	0.434835
2	Value_vehicle	0.433717
3	weight_log	0.392169
4	value_per_vehicle_year	0.355450
5	N_doors	0.346052
6	Year_matriculation	0.327107
7	vehicle_age	-0.326919
8	Weight	0.298471
9	Cylinder_capacity	0.289776
10	length_log	0.209474
11	Length	0.205979
12	age_at_renewal	-0.115744
13	licence_tenure	-0.113998
14	engine_cc_per_power	-0.107788

Table 9. Main correlations with Premium

Feature	Pearson Correlation with Premium
vehicle_value_log	0.457
Power	0.435
Value_vehicle	0.434
weight_log	0.392
value_per_vehicle_year	0.355
N_doors	0.346
Year_matriculation	0.327
vehicle_age	-0.327
Weight	0.298
Cylinder_capacity	0.290

The correlation heatmap reveals high redundancy among some numerical features, such as raw and log-transformed versions of vehicle measurements, as well as vehicle year and vehicle age. This finding influences the modeling design. Linear models use a more compact feature view to reduce instability from multicollinearity, while tree-based models use a fuller feature view because tree models can handle redundant variables more flexibly.



3.6 Preprocessing Pipeline Design

The project uses model-specific preprocessing pipelines. This design ensures that every model receives appropriate transformations while preventing data leakage.

For linear models, the project uses a compact feature view with 35 columns. Numerical variables are imputed using the median and standardized. Categorical variables are imputed with a constant Missing category and one-hot encoded. Binary flags are passed through as 0/1 indicators. Standardization is necessary because regularized linear models are sensitive to feature scale.

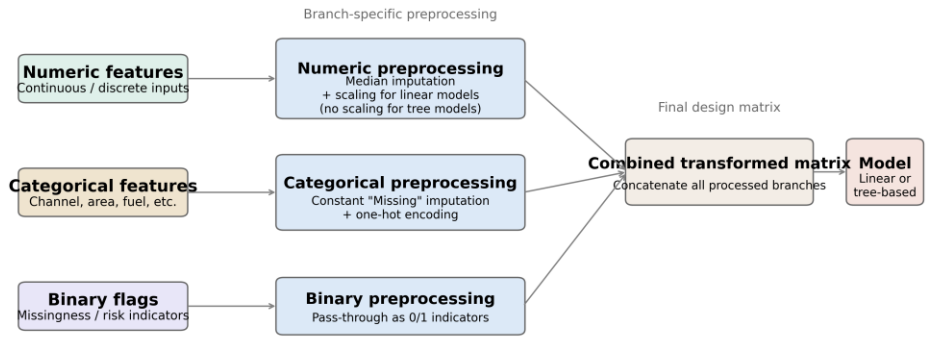
For tree-based models, the project uses the full 41-feature view. Numerical variables are median-imputed but not standardized, because tree-based models split variables by thresholds and do not require scaling. Categorical variables are imputed and one-hot encoded. Binary flags are retained directly.

After preprocessing, the transformed feature matrices contain 39 columns for linear models and 48 columns for tree-based models.

Table 10. Preprocessing design by model family

Model Family	Feature View	Numeric Processing	Categorical Processing	Binary Flag Processing	Transformed Feature Count
Linear models	35 selected features	Median imputation + standard scaling	Constant Missing imputation + one-hot encoding	Passed through as 0/1	39
Tree-based models	41 full features	Median imputation	Constant Missing imputation + one-hot encoding	Passed through as 0/1	48

Figure 8. Preprocessing Pipeline Diagram



The use of pipelines is important because imputation, scaling, and encoding are learned only from the training data. The same fitted transformations are then applied to validation and test data. This prevents future information from entering the preprocessing stage and makes the workflow reproducible.

4. Modeling Strategy and Results

4.1 Model Families and Target Scales

The modeling strategy compares both linear and tree-based regression models.

The linear models include ordinary least squares, Ridge, Lasso, and Elastic Net. These models provide interpretable baselines and test whether the engineered features can explain premium through an approximately linear relationship.

The tree-based models include Random Forest, Extra Trees, Gradient Boosting, and Hist Gradient Boosting. These models are able to capture nonlinear effects, threshold behavior, and interactions among variables. This is especially relevant in motor insurance, where the effect of vehicle value may depend on vehicle age, and the effect of driver profile may interact with vehicle characteristics and contract structure.

The study also compares two target scales: raw Premium and log1p(Premium). The log transformation is motivated by the right-skewed distribution of premium in the training set. It reduces the influence of extreme high-premium records and may improve average prediction error.

4.2 Benchmark Modeling

Benchmark modeling is used to establish the first comparison across model families and target scales. The purpose is not to select the final model immediately, but to understand whether the pricing relationship is closer to a linear structure or requires nonlinear models.

The benchmark results show that tree-based models clearly outperform linear models. Extra Trees achieves the strongest validation MAE in the initial benchmark, followed by Random Forest. Linear models perform worse, which suggests that premium prediction depends on nonlinear relationships and interactions that are not fully captured by additive linear models.

The comparison between raw and log-transformed targets shows that log1p(Premium) improves validation MAE for several models. For example, Ridge improves from 78.39 to 74.36 MAE, Random Forest improves from 62.30 to 59.83 MAE, and Extra Trees improves from 55.29 to 52.78 MAE. This confirms that the target distribution affects model performance.

At the same time, the benchmark stage reveals overfitting risk. Extra Trees produces nearly zero training error but a much higher validation error. This means that the model can memorize the training data if it is not properly regularized.

Therefore, the benchmark results are used to identify promising model families, while final model selection is based on chronological validation and tuning.

Table 11. Benchmark validation comparison

Model	Target Scale	Train MAE	Validation MAE	Train RMSE	Validation RMSE	Train R²	Validation R²
Extra Trees	Premium	0.00	55.29	0.01	89.37	1.000	0.606
Extra Trees	log1p(Premium)	0.00	52.78	0.01	89.76	1.000	0.602
Random Forest	Premium	21.30	62.30	34.88	97.52	0.939	0.531
Random Forest	log1p(Premium)	21.57	59.83	40.79	98.65	0.916	0.520
Hist Gradient Boosting	Premium	60.29	69.76	91.53	104.64	0.579	0.460
Hist Gradient Boosting	log1p(Premium)	59.53	67.44	96.40	105.98	0.533	0.446
Gradient Boosting	Premium	64.76	71.72	100.51	109.11	0.492	0.412
Gradient Boosting	log1p(Premium)	63.21	69.19	103.19	109.62	0.465	0.407
Ridge	log1p(Premium)	68.70	74.36	110.85	115.23	0.382	0.345
OLS	log1p(Premium)	68.70	74.36	110.85	115.23	0.382	0.345

Table 12. Raw versus log target comparison

Model	Raw Validation MAE	Log Validation MAE	MAE Improvement from Log Target
OLS	78.39	74.36	4.03
Ridge	78.39	74.36	4.03
Extra Trees	55.29	52.78	2.51
Gradient Boosting	71.72	69.19	2.54
Random Forest	62.30	59.83	2.47
Hist Gradient Boosting	69.76	67.44	2.32
Elastic Net	82.21	93.20	-10.99
Lasso	78.96	93.20	-14.24

4.3 Chronological Cross-Validation and Hyperparameter Tuning

After benchmark modeling, the project performs hyperparameter tuning using chronological cross-validation. This step is necessary because a single validation split may not fully represent model stability over time. The tuning process uses earlier observations for training and later observations for validation within the training period, preserving the time direction of the prediction task.

The tuning stage focuses on promising tree-based models: Extra Trees, Random Forest, and Hist Gradient Boosting. Each model is evaluated under both raw and log-transformed target settings. Random search is used to explore hyperparameter combinations efficiently.

The tuned validation results show that Random Forest with log1p(Premium) achieves the best validation MAE among the final tuned candidates. Although Extra Trees performs strongly, Random Forest with the log-transformed target provides the most favorable validation MAE and is selected as the final primary model.

Table 13. Tuned model comparison on validation set

Model	Target Scale	Train MAE	Validation MAE	MAE Gap	Train RMSE	Validation RMSE	Train R²	Validation R²
Random Forest	log1p(Premium)	15.70	56.93	41.24	36.82	96.55	0.932	0.540
Extra Trees	Premium	34.90	62.23	27.34	57.06	96.85	0.836	0.537
Extra Trees	log1p(Premium)	34.59	59.98	25.39	61.70	97.86	0.809	0.527
Random Forest	Premium	35.30	62.31	27.01	61.17	98.64	0.812	0.520
Hist Gradient Boosting	log1p(Premium)	51.09	64.47	13.38	82.70	100.97	0.656	0.497
Hist Gradient Boosting	Premium	60.19	69.33	9.14	93.01	104.55	0.565	0.460

The selected model has a relatively large training-validation gap, which indicates that some overfitting remains. However, among the tuned candidates it provides the best validation MAE. Therefore, the final decision is based on both validation performance and subsequent out-of-time test evaluation.

4.4 Final Out-of-Time Test Evaluation

After model selection, the final candidate models are retrained using the combined training and validation data, and then evaluated on the out-of-time test set. The test set is not used for feature engineering, benchmark comparison, tuning, or model selection. It represents the final evaluation of forward-time predictive performance.

The final selected model is Random Forest trained on log1p(Premium). It outperforms the Extra Trees candidate on all three test metrics: MAE, RMSE, and R^2.

Table 14. Final out-of-time test results

Model	Target Scale	Train Full MAE	Test MAE	Train Full RMSE	Test RMSE	Train Full R^2	Test R^2
Random Forest	log1p(Premium)	15.39	55.08	36.51	91.05	0.933	0.573
Extra Trees	Premium	35.12	60.27	57.02	92.55	0.837	0.559

A test MAE of 55.08 means that the average absolute prediction error is approximately 55 premium units. The test RMSE of 91.05 is larger than the MAE, which indicates that some records have relatively large prediction errors. The test R^2 of 0.573 means that the final model explains approximately 57.3% of the variation in the out-of-time test premiums.

The test residual summary shows that the model's average prediction is close to the average actual premium. The mean actual premium in the test set is 310.71, while the mean predicted premium is 305.66. The mean residual is 5.05, which suggests that the model is not strongly biased on average. However, the residual range is wide, which indicates that some individual policies remain difficult to predict.

The error-by-bucket analysis shows that model performance differs across premium levels. The model performs best in the middle premium ranges, while the highest premium bucket is the most difficult to predict. In the highest bucket, the actual mean premium is 519.25, but the predicted mean is 411.08, and the MAE rises to 121.34. This suggests that the model tends to underpredict some high-premium policies.

Table 15. Test error by actual premium bucket

Actual Premium Bucket	Observations	Actual Mean	Predicted Mean	MAE	RMSE
40.20 to 222.15	5,039	158.81	197.34	45.26	70.42
222.15 to 266.44	5,040	245.47	277.25	38.20	53.82
266.44 to 310.56	5,038	287.36	305.17	32.61	49.54
310.56 to 385.38	5,038	342.67	337.46	37.99	53.81
385.38 to 2116.69	5,039	519.25	411.08	121.34	168.05

This pattern is important for interpretation. High-premium policies may involve special underwriting rules, rare vehicle types, discounts, coverage differences, or commercial pricing decisions that are not fully captured by the available variables.

5. Model Interpretation

5.1 Feature Importance Analysis

Model interpretation is essential for insurance applications. A premium prediction model should not only provide accurate predictions, but also show whether its predictions are driven by reasonable insurance-related variables.

The first interpretation method used in this study is permutation importance. Permutation importance measures how much model performance worsens when a feature is randomly shuffled. If shuffling a feature leads to a large increase in prediction error, the feature is important for the model.

The permutation importance results show that the most important features include value_per_vehicle_year, Payment, flag_n_doors_zero, policy_tenure, Value_vehicle, vehicle_value_log, licence_tenure, Cylinder_capacity, age_at_renewal, and N_doors.

These results are consistent with the insurance context. Vehicle value intensity, payment method, special vehicle indicators, contract duration, vehicle economic value, and driver experience all have plausible relationships with premium. The strong importance of flag_n_doors_zero indicates that unusual vehicle records are not random noise. Instead, they form a distinct pattern that the model uses in prediction.

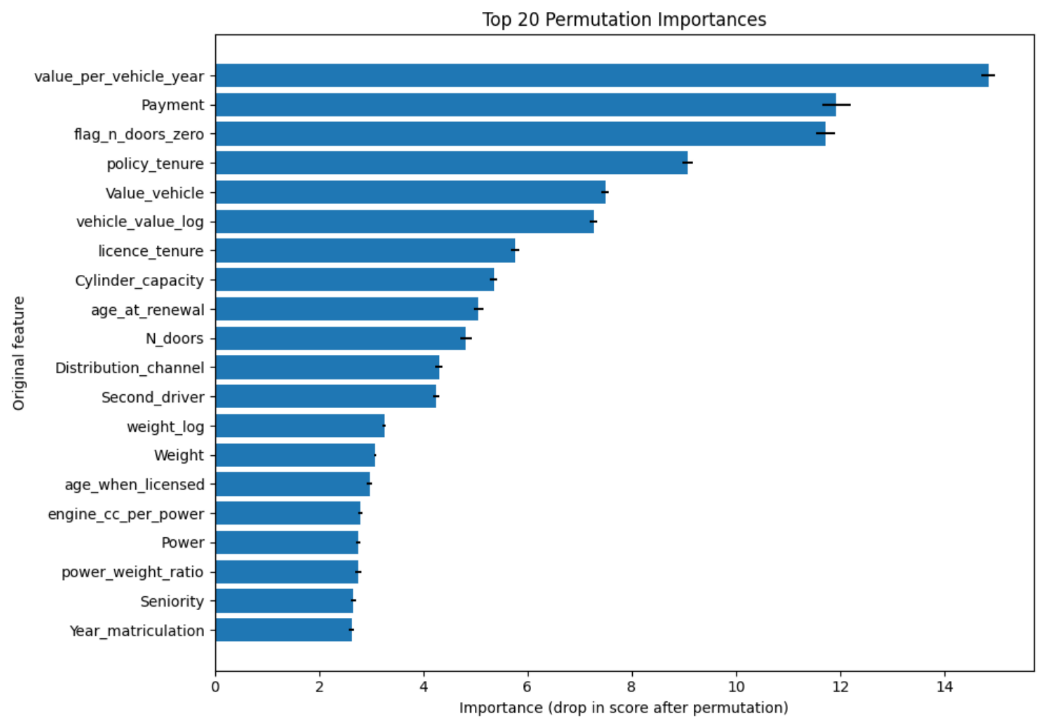


Table 16. Top permutation importance features

Rank	Feature	Importance Mean	Importance Std.
1	value_per_vehicle_year	14.85	0.13
2	Payment	11.93	0.27
3	flag_n_doors_zero	11.72	0.18
4	policy_tenure	9.07	0.11
5	Value_vehicle	7.49	0.07
6	vehicle_value_log	7.27	0.07
7	licence_tenure	5.76	0.08
8	Cylinder_capacity	5.35	0.07
9	age_at_renewal	5.06	0.09
10	N_doors	4.82	0.12
11	Distribution_channel	4.30	0.07
12	Second_driver	4.25	0.07
13	weight_log	3.25	0.04
14	Weight	3.07	0.03
15	age_when_licensed	2.97	0.05

5.2 SHAP Methodology

Permutation importance provides a global ranking of predictive dependence, but it does not explain the direction or local contribution of each feature. Therefore, this study also uses SHAP values.

SHAP, or SHapley Additive exPlanations, decomposes a model prediction into a baseline value and a set of feature-level contributions. For a given policy, the model prediction can be interpreted as a baseline value plus positive and negative contributions from individual variables.

SHAP is useful for this project because it provides both global and local interpretability. Globally, it shows which variables are most important across many test samples. Locally, it explains why a specific policy receives a high or low predicted premium.

Because the final model is a Random Forest, SHAP is well suited for analyzing nonlinear feature effects and interactions. In this project, SHAP is applied to a sample of transformed test observations and then grouped back to the original feature level to make the explanation more interpretable.

Table 17. Interpretation of SHAP components

Component	Meaning
Baseline value	The model's expected prediction before considering individual feature values
Positive SHAP value	The feature pushes the predicted premium upward
Negative SHAP value	The feature pushes the predicted premium downward
Larger absolute SHAP value	The feature has a stronger influence on that prediction

5.3 SHAP Results and Business Interpretation

The grouped SHAP results are broadly consistent with the permutation importance results. The most important original features include flag_n_doors_zero, Payment, value_per_vehicle_year, policy_tenure, N_doors, Value_vehicle, vehicle_value_log, Second_driver, licence_tenure, and Distribution_channel.

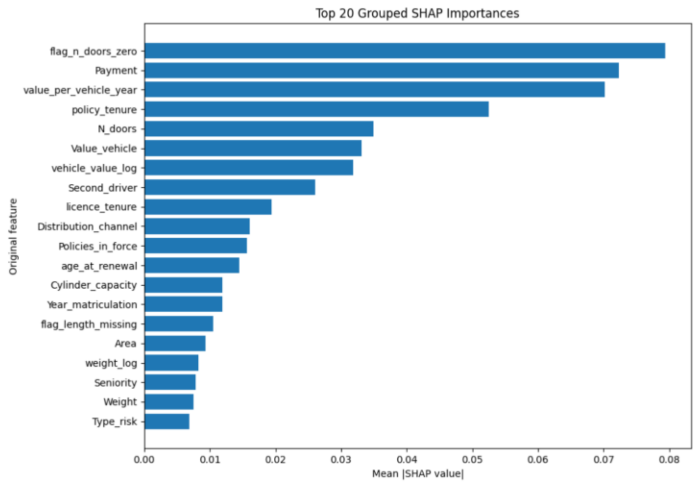


Table 18. Grouped SHAP importance

Rank	Original Feature	Mean Absolute SHAP Value
1	flag_n_doors_zero	0.0794
2	Payment	0.0723
3	value_per_vehicle_year	0.0702
4	policy_tenure	0.0525
5	N_doors	0.0350
6	Value_vehicle	0.0331
7	vehicle_value_log	0.0319
8	Second_driver	0.0261
9	licence_tenure	0.0194
10	Distribution_channel	0.0161
11	Policies_in_force	0.0157
12	age_at_renewal	0.0145
13	Cylinder_capacity	0.0119
14	Year_matriculation	0.0118
15	flag_length_missing	0.0105

The interpretation can be grouped into four main dimensions.

First, vehicle value and value intensity are central. Value_vehicle, vehicle_value_log, and value_per_vehicle_year all appear among the important features. This indicates that the model learns both the absolute value of the insured vehicle and the value relative to the vehicle's age. In insurance pricing, this is reasonable because higher-value vehicles may lead to higher potential claim costs.

Second, vehicle structure is important. N_doors, flag_n_doors_zero, Cylinder_capacity, Weight, and related vehicle features contribute to predictions. The strong effect of flag_n_doors_zero suggests that unusual vehicle structure or special coding patterns identify a distinct pricing group. This confirms the value of preserving abnormality indicators rather than deleting special records.

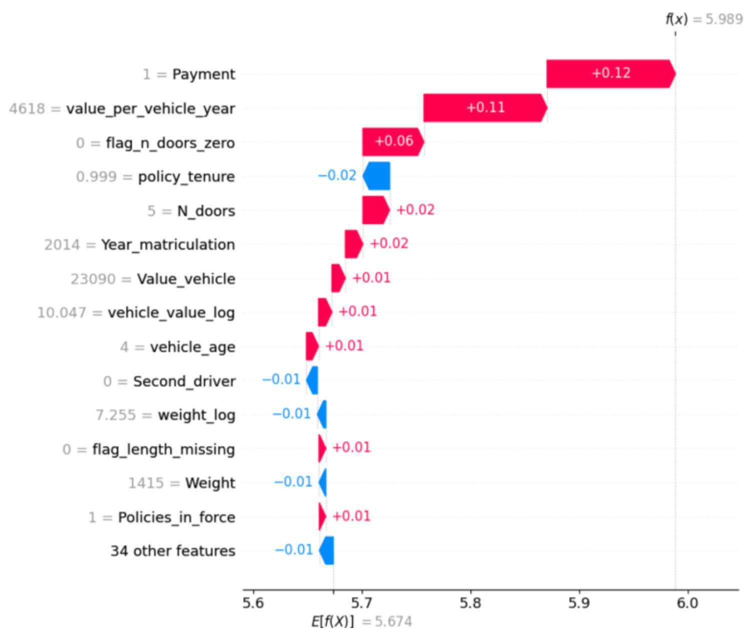
Third, contract and payment information matter. Payment, policy_tenure, Policies_in_force, Distribution_channel, and Second_driver are all relevant. These variables may capture differences in policy structure, customer relationship, sales channel, and business segmentation. Premium is therefore not determined only by vehicle risk; it also reflects contract and customer-policy characteristics.

Fourth, driver profile contributes to prediction. licence_tenure, age_at_renewal, and age_when_licensed appear in the importance analyses. This supports the insurance intuition that driving experience and age are relevant to premium formation, even when vehicle variables are dominant.

5.4 Local Error Interpretation with SHAP

In addition to global interpretation, this study uses SHAP waterfall plots to examine individual prediction errors. This helps explain where the model succeeds and where it struggles.

The first case is a high-premium underprediction. The actual premium is 1589.16, while the model predicts 397.87, producing a residual of -1191.29. This is a major underprediction. The policy has a vehicle value of 23,090, vehicle age of 4 years, value per vehicle year of 4,618, and licence tenure of 44.01 years. These features do not fully explain the extremely high observed premium, suggesting that important pricing factors may be missing from the dataset. Possible explanations include special coverage, underwriting adjustments, rare policy conditions, or other business rules not represented in the available variables.



The second case is a low-premium overprediction. The actual premium is 132.16, while the model predicts 697.47, producing a residual of 565.31. This policy has a very high vehicle value of 155,252.80 and value per vehicle year of 11,089.49. The model therefore predicts a high premium based on vehicle economic value, but the actual premium is much lower. This may reflect discounts, special product design, commercial pricing strategy, or policy-specific rules that are not observed in the modeling data.

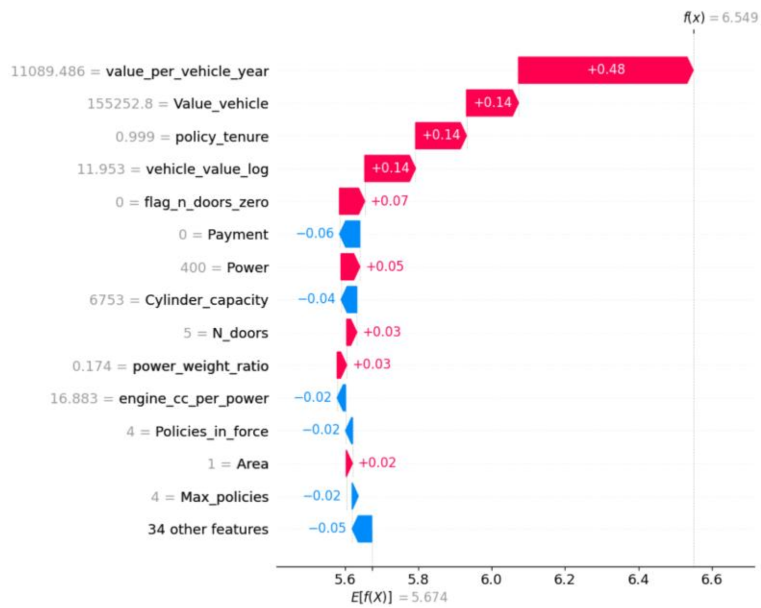


Table 19. Local error case summary

Case	Actual Premium	Predicted Premium	Residual	Key Observed Characteristics	Interpretation
High-premium underprediction	1589.16	397.87	-1191.29	Vehicle value 23,090; vehicle age 4; value per vehicle year 4,618; licence tenure 44.01	Available features do not fully explain the unusually high premium
Low-premium overprediction	132.16	697.47	565.31	Vehicle value 155,252.80; vehicle age 13; value per vehicle year 11,089.49; licence tenure 37.61	Model predicts high premium from vehicle value, but actual premium may reflect discounts or special pricing rules

These local cases highlight an important limitation of observed premium modeling. The target variable is not pure technical risk cost. It may also include discounts, commercial adjustments, customer retention decisions, and underwriting rules that are not fully included in the dataset. Therefore, even a strong model may struggle with policies whose premiums are heavily influenced by unobserved business factors.

6. Conclusion and Insurance Market Implications

This study develops an ex-ante machine learning framework for motor insurance premium prediction. It begins with a careful understanding of the original variables and excludes information that may not be available at the renewal pricing date. Then it constructs business-driven features from customer, contract, and vehicle information, applies chronological validation, compares linear and tree-based models, tunes candidate models, evaluates the final model on an out-of-time test set, and interprets the model using permutation importance and SHAP.

The final selected model, Random Forest with log1p(Premium), provides the best out-of-time performance among the final candidates. More importantly, the model's behavior is interpretable. The major predictive drivers are related to vehicle value, vehicle structure, payment method, policy tenure, distribution channel, and driving experience. These variables are consistent with real insurance pricing intuition.

The contribution of this project to the insurance market is threefold. First, it demonstrates how machine learning can be applied to premium prediction while respecting the information available at pricing time. Second, it shows that business-driven feature engineering can translate raw insurance data into meaningful pricing signals. Third, it illustrates how model interpretation can help connect predictive performance with business understanding.

7. Limitations and Future Work

1. the target variable is observed premium, not pure claim cost or technical premium. Observed premium may include discounts, business strategy, expenses, market competition, and customer retention effects. Therefore, the model learns historical pricing behavior rather than a pure actuarial risk function.

2. potentially useful claim-related variables are excluded or treated cautiously because their timing is unclear or they may contain future information. This conservative choice improves the validity of the ex-ante design, but it may reduce predictive performance.

3. high-premium policies are more difficult to predict. The error-by-bucket analysis shows that the highest premium group has the largest MAE. This suggests that tail-risk modeling, high-premium segmentation, or quantile-based methods may be useful in future work.

4. the current validation design uses a chronological train-validation-test split and inner chronological validation. Future research could extend this to a fuller rolling walk-forward backtesting framework to evaluate stability across multiple future periods.

5. the model can be extended with additional algorithms. Potential future models include LightGBM, XGBoost, CatBoost, Tweedie regression, generalized additive models, and quantile regression. These methods may improve performance, especially for nonlinear effects and high-premium tail cases.

Finally, future work can expand the interpretability analysis by segment. SHAP effects can be compared across vehicle types, payment methods, risk areas, customer age groups, and premium bands. This would help determine whether the model behaves consistently across different insurance market segments.